AML Project Ideas: 7 categories (Group of 3 maximum)

A. Core Theoretical ML Projects

1. Bias–Variance Decomposition in Practice

- Compare:
  - kNN, Decision Trees, Linear Models, SVM
- Experiment:
  - Vary dataset size + noise
- Research question:

How does bias–variance tradeoff behave across models?

Bias/variance is a central ML concept tied to generalization

2. Regularization Study (L1 vs L2 vs Elastic Net)

- Models: Linear / Logistic Regression
- Analyze:
  - Feature selection behavior (L1 sparsity)
  - Stability vs overfitting
- Add:
  - Correlated features experiment

3. Optimization Algorithms Comparison

- Compare:
  - Gradient Descent, SGD, Mini-batch
- Study:
  - Convergence speed
  - Stability under noise

Optimization is a core ML research direction with many open problems

B. Feature Engineering & Data-Centric ML

Classical ML heavily depends on feature engineering

4. Impact of Feature Engineering on Model Performance

- Compare:
    - Raw features vs engineered features
- Techniques:
    - Polynomial features
    - Interaction terms
    - Scaling
- Research question:

Can feature engineering outperform complex models?

5. Automatic Feature Selection vs Manual Design

- Compare:
    - PCA
    - SelectKBest
    - Recursive Feature Elimination
- Evaluate:
    - Accuracy vs interpretability

6. Dimensionality Reduction Study

- Compare:
    - PCA vs LDA vs feature selection
- Analyze:
    - Accuracy vs computation time
    - Visualization quality

C. Classical Algorithms Deep Dive

7. kNN vs SVM vs Decision Trees (Systematic Study)

- Evaluate:
    - Sensitivity to scaling
    - Curse of dimensionality
- Add:
    - Theoretical explanation

8. Kernel Methods Study (SVM)

- Compare kernels:
    - Linear vs Polynomial vs RBF
- Research:

When does kernel trick really help?

9. Ensemble Learning Investigation

- Compare:
    - Bagging (Random Forest)
    - Boosting (AdaBoost, Gradient Boosting)
- Analyze:
    - Variance reduction vs bias

D. Unsupervised Learning

Unsupervised learning discovers structure in unlabeled data

10. Clustering Stability Analysis

- Compare:
    - k-means, DBSCAN, hierarchical clustering
- Study:
    - Sensitivity to initialization
    - Cluster validity metrics

11. Anomaly Detection Study

- Methods:
    - Isolation Forest
    - One-class SVM
    - Statistical methods
- Dataset:
    - Fraud / network logs

12. Clustering vs Classification Relationship

- Use clustering to:
    - Generate labels
- Then:
    - Train supervised model
- Research:

When can unsupervised learning replace labels?

E. Robustness, Noise & Real-World ML

13. Effect of Noisy Data on ML Algorithms

- Inject noise:
    - Label noise
    - Feature noise
- Compare robustness across models

14. Missing Data Handling Strategies

- Compare:
    - Mean imputation
    - KNN imputation
    - Model-based imputation
- Evaluate:
    - Impact on accuracy

15. Outlier Detection and Its Effect on Models

- Compare:
    - With vs without outlier removal
- Study:
    - Sensitivity of each algorithm

F. Advanced

16. Hybrid Rule-Based + ML Systems

- Combine:
    - Domain rules + ML model
- Inspired by rule-based ML systems
- Research:

Does adding domain knowledge improve performance?

17. Meta-Learning for Algorithm Selection

- Build a system that:
    - Predicts best ML algorithm for dataset
- Features:
    - Dataset size, variance, sparsity

18. Cost-Sensitive Learning Study

- Modify loss:
    - Penalize certain errors more
- Application:
    - Medical / fraud

19. Model Interpretability vs Accuracy

- Compare:
    - Logistic Regression vs Random Forest
- Measure:
    - Explainability + performance trade-off

20. Data Drift & Model Degradation

- Simulate:

    - Changing data distribution
- Study:
    - When models fail

G. Applied

21. Credit Risk Prediction

- Models:
    - Logistic Regression vs Tree-based
- Add:
    - Explainability

22. Student Performance Prediction

- Features:
    - Attendance, grades, engagement
- Add:
    - Feature importance analysis

23. Customer Segmentation (Clustering)

- Compare clustering algorithms
- Add:
    - Business interpretation

General recommendations:

Add rigor:

- **Statistical tests (t-test, cross-validation)**
- **Confidence intervals**

Add theory:

- **Derive formulas (loss, margin, bias-variance)**

Add experiments:

- **Synthetic + real datasets**

Add comparison:

- **≥ 3 algorithms**

Rubrics for assessment

- 7 main criteria
- 4 performance levels
- Designed for research + implementation + analysis

1. Problem Definition & Research Motivation (10 pts)

Level	Description
Excellent (9–10)	Clear, well-defined ML problem with strong motivation; grounded in real-world or research gap
Good (7–8)	Problem clear but limited justification or novelty
Fair (5–6)	Vague problem or weak relevance
Poor (0–4)	Unclear or trivial problem

2. Literature Review & Theoretical Understanding (15 pts)

Level	Description
Excellent (13–15)	Strong review of related work; identifies gaps; connects theory (bias-variance, regularization, etc.) , at least 15 references
Good (10–12)	Adequate references; some theoretical grounding
Fair (7–9)	Minimal literature; weak theoretical links
Poor (0–6)	No meaningful literature or theory

3. Methodology & Experimental Design (15 pts)

Level	Description
Excellent (17–20)	Well-justified model choices; multiple algorithms; controlled experiments; clear pipeline
Good (13–16)	Good design but limited comparison or justification
Fair (9–12)	Basic implementation; weak experimental design

Level	Description
Poor (0–8)	Ad hoc or incorrect methodology

Should include:

- Model selection (e.g., SVM, kNN, Logistic Regression)
- Hyperparameter tuning
- Cross-validation

4. Implementation & Code Quality (15 pts)

Level	Description
Excellent (13–15)	Clean, modular, reproducible code; proper pipeline
Good (10–12)	Functional but lacks structure
Fair (7–9)	Poor organization; limited reproducibility
Poor (0–6)	Broken / incomplete implementation

5. Evaluation, Results & Analysis (20 pts)

Level	Description
Excellent (22–25)	Deep evaluation using multiple metrics (precision, recall, etc.), cross-validation, error analysis, statistical insight
Good (18–21)	Good evaluation but limited depth
Fair (13–17)	Basic metrics only (e.g., accuracy only)
Poor (0–12)	Weak or incorrect evaluation

High-level expectations:

- Compare ≥ 3 models
- Use multiple metrics
- Analyze why results occur

Strong projects:

- Check overfitting

- Perform sensitivity analysis

6. Critical Thinking & Research Insight (10 pts)

Level	Description
Excellent (9–10)	Deep insights, limitations, future work, strong reasoning
Good (7–8)	Some interpretation and discussion
Fair (5–6)	Superficial conclusions
Poor (0–4)	No meaningful analysis

7. Report & Presentation (15 pts). Report to be paper style between 10 and 15 pages

Level	Description
Excellent (5)	Clear, structured, professional (paper-style)
Good (4)	Mostly clear
Fair (3)	Some clarity issues
Poor (0–2)	Poor communication

Final Weight Summary

Component	Weight
Problem Definition	10
Literature Review	15
Methodology	20
Implementation	15
Evaluation & Analysis	25
Critical Thinking	10
Report	5